

P5: Project Report

RunPal

Tag Line: A smartwatch that allows for runners to run along with a digital pet. Can be controlled handsfree with the user's voice.

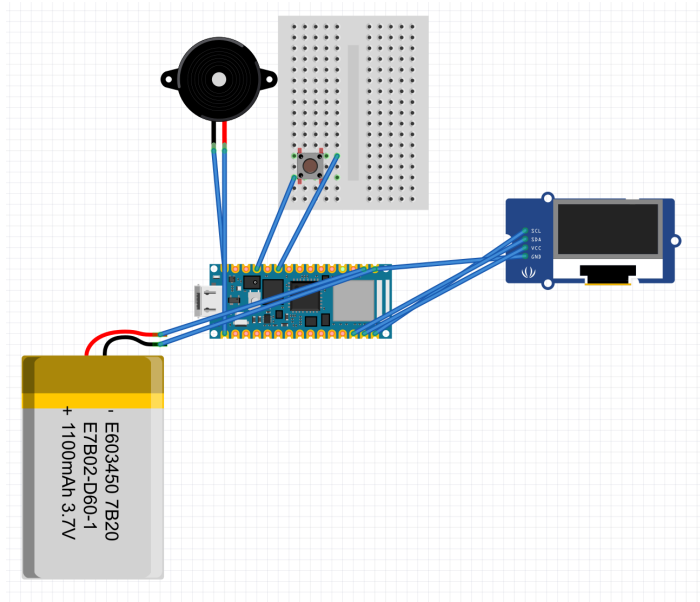
Description: The purpose of this product is to motivate runners with a digital pet as runners require a motivation to reach their running goals. Runners have also identified that they want to be able to use running tools without having to use their hands, from previous research. This product can accomplish this by allowing users to start a timer for their run with their voice, as the timer starts there is a pet that runs along with the runner, working as a motivation for the runner.

Diagram:

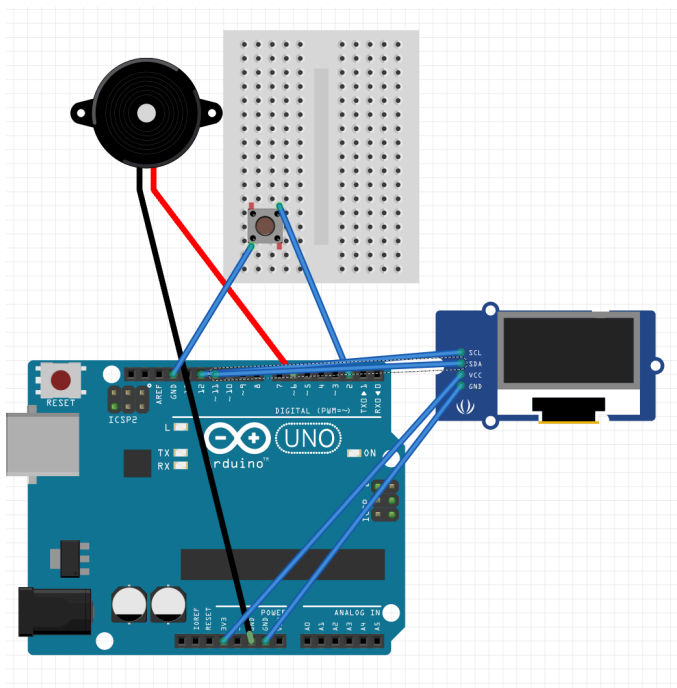


Implementation: Utilizing Edge Impulse along with the nano's TinyML and built-in microphone, I am able to train a model to recognize two voice commands. The first command simply starts the timer for the runner by saying, "Hey Ajax start a run". This command can be changed to the user's pet name but the model would have to be retrained. The second command is much simpler with it being able to end the run with, "stop run". Using these commands the user is able to control the device handsfree. The user is also able to view the timer and pet through the OLED display which helps with motivating the runner. The buzzer serves as a way to notify the runner that their timer is starting or ending. Lastly, the button serves as an alternative to the voice commands and can also start and end runs.

Circuit Diagram:



Circuit for the nano version



Circuit for the uno version

Software: The software for RunPal is structured around a finite state machine with five states: IDLE, COUNTDOWN, RUNNING, STOPPING, and SUMMARY. Each state controls what is displayed on the OLED, what inputs are accepted, and what sounds are played at any given moment.

In the IDLE state the cat sits and blinks while the system continuously listens for voice input through the onboard PDM microphone. Audio is sampled in real time and passed through a TFLite machine learning model trained with Edge Impulse to recognize two keywords: "hey ajax" and "stop run". When "hey ajax" is detected above a confidence threshold the system transitions to COUNTDOWN, which runs a 5 second timer and displays a large countdown number on screen. Once the countdown reaches zero the state advances to RUNNING, the buzzer plays three rising beeps, and the cat animation begins.

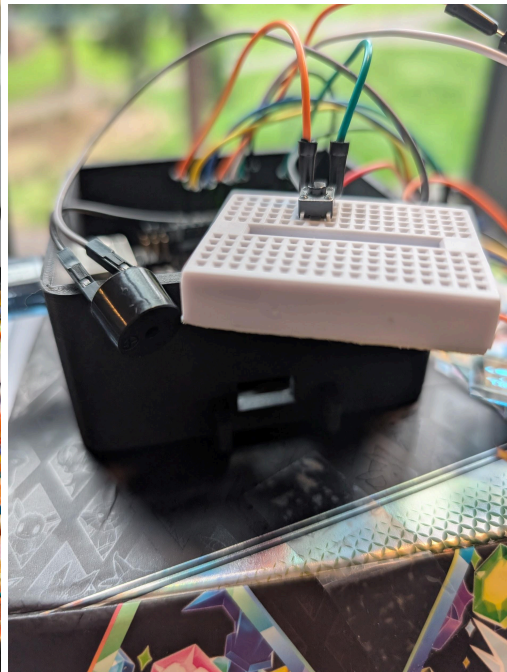
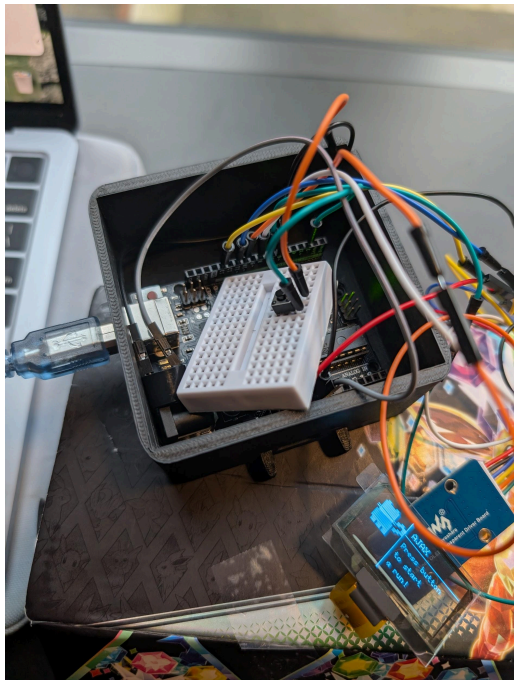
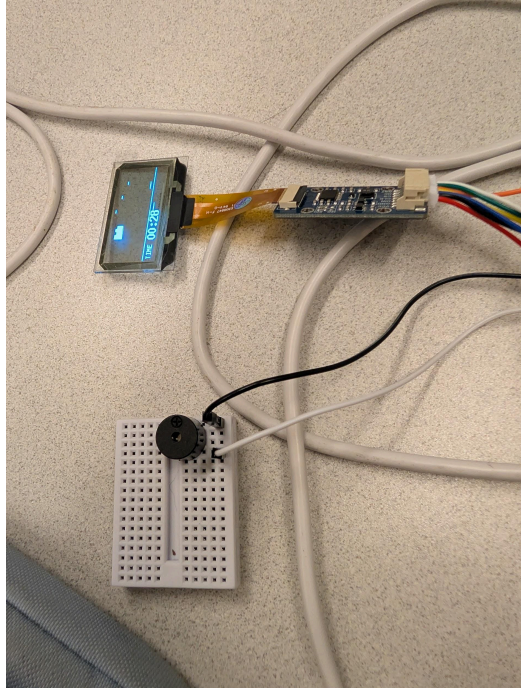
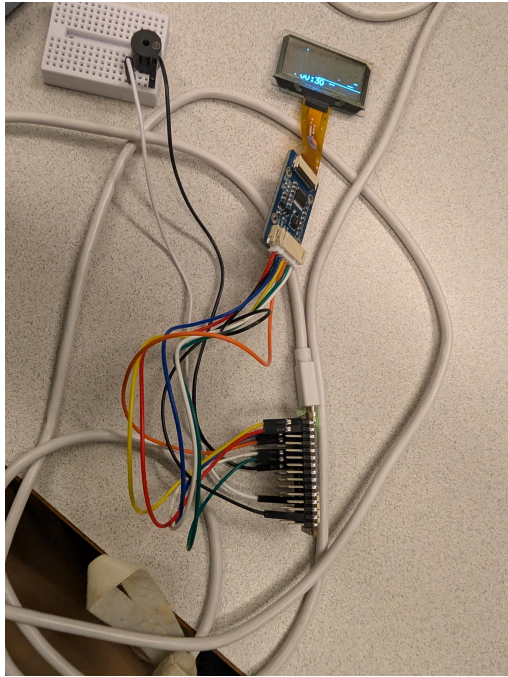
While in the RUNNING state the elapsed time is calculated every loop and displayed at the bottom of the screen. The microphone continues listening and when "stop run" is detected the timer is frozen at that exact millisecond, the buzzer plays two descending beeps, and the system moves to STOPPING. The button also serves as a manual stop in case voice recognition is unavailable. The STOPPING state displays a brief pause screen before automatically advancing to SUMMARY, which shows the final frozen run time. After 10 seconds the system resets to IDLE and resumes listening.

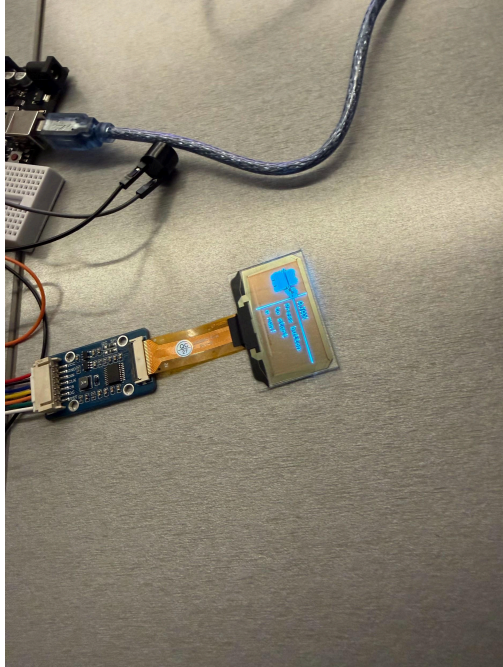
The display rendering is handled entirely with the U8g2 library using a pixel art drawing approach — all cat graphics are built from filled rectangles on a 2×2 pixel grid to achieve a chunky 8-bit aesthetic. The running animation cycles through 4 frames with bounce physics, speed lines, and a tail wag. The idle cat has a blink cycle with a happy eye expression after reopening.

Audio inference runs on every loop cycle during IDLE and RUNNING states only. During COUNTDOWN, STOPPING and SUMMARY the inference is paused and a fixed delay is used instead to keep animations smooth. Button input uses software debouncing to prevent false triggers. The main loop runs continuously, updating state, drawing the frame buffer, running inference, and sending the completed frame to the display on each cycle.

The final version using the uno omits the voice commands and just uses the button.

Gallery:





Images of both the nano and uno version

Process: The process for this project was fairly simple. I began by first getting the OLED screen to display what I wanted by using Claude to tell me what libraries I may need at first. Then through a few prompts I was able to display a cat. I realized at that point that the screen I had would not work for my concept of running glasses and pivoted to a smartwatch. I then utilized Claude to guide me on how to train the model and worked on using Edge Impulse to record and train it while Claude provided the right libraries and wrote the code to implement the model. From there I connected the nano to the display and also integrated the button and had Claude revise the code to make the display show the cat running and the timer when using those inputs along with a buzzer to output sound. The next step was to use Claude to design a watch case using the dimensions of the components. It produced a OpenSCAD file that I could use in OpenSCAD and then import into Rhino. In Rhino I then edited the dimensions a bit and moved a few components around and exported it to the Prusa slicer so I could get the gcode to 3d print it.

The last step would have been to just put everything together inside the case and attach a wrist band but during the assembly process the battery I had used fried the nano. I'm not quite sure what exactly destroyed the nano as I had originally tested it with the battery before trying to assemble it and it worked fine. The issue might have been that during the last steps I utilized an adapter so that I could have more ground inputs and

when I plugged in the battery the buzzer started making a high-pitched noise and then when I took it all apart to just test it by itself the board was not working anymore. Utilizing Claude I modified the code to work with the Uno but that meant losing the ability to use the voice commands and it not being able to fit properly inside my case.

I think the takeaways I had from this was to check that all the components are compatible. I was having trouble finding the right battery that would work with my nano and needed one that would arrive quickly. Additionally, making sure I am managing my time is also important as most of the steps I took in the process were taken last minute.